GEN 1.7 Differences from ICAO Standards, Recommended Practices and Procedures

A list of significant differences between national regulations and practices of the State and related ICAO provisions, including:

1) provision affected (Annex as amended, paragraph); and
2) difference in full text.

1 Annex 1 - Personnel Licensing

CHAPTER 1

Paragraph reference: 1.2.

1.2.1
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011, 61.01.02(1) states that no person may act as a pilot of a South African registered aircraft, except in the case of dual instruction with an appropriately rated flight instructor, unless such person holds a valid pilot licence with applicable ratings issued, revalidated or re issued by the Director or by an appropriate authority validated by the Director in terms of The Civil Aviation Regulations, 2011, Part 61 or Part 62: Provided that a student pilot licence may have been issued without a class rating or type rating.

1.2.1.1
South Africa has not implemented a multi-crew pilot and flight navigator license.

1.2.2.2
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011, 61.01.13(4) states that before the Director validates or converts a foreign licence or rating for a commercial air transport operation or a PPL with Instrument Rating (PPL/IR), he or she must confirm the validity of the foreign licence or rating with the appropriate authority of the issuing Contracting State.

1.2.2.3
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011, 61.01.13(5) requires that certain theoretical knowledge examinations and a skills tests are required before a licence may be validated.

1.2.5.2.2
South Africa's regulation is less exacting in that the South African Civil Aviation Regulations, 2011, 67.00.6 (1)(d) requires that commercial pilots over the age of 40

years, in a single pilot operation do not have their medical assessment validity reduced to 6 months unless there is a risk factor where 6 monthly reports must be issued.

1.2.5.2.4
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011, 67.00.6 (3)(b) requires that the holders of Class 2 and 3 medical certificates who have passed their 40th birthday have the period of validity for their medical assessment reduced to 12 months and 67.00.6 (4)(b) states that the period of validity of Class 4 medical certificates are reduced to 36 months when the pilot has passed his or her 40th birthday.

1.2.5.2.5
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011, 67.00.6 (3)(b) requires that the holders of Class 2 and 3 medical certificates who are over 40 years of age are required to submit a medical every twelve months. There is currently no such restriction on the holder of a Class 4 medical certificate, however, this will be amended during 2013 to fall in line with the previous provision.

1.2.9.2
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011, 65.01.8 requires that Air traffic service personnel shall have sufficient ability in reading, speaking and understanding the English language to enable them to adequately carry out their responsibilities as air traffic service personnel and have attained a minimum of ICAO level 4 in their English language proficiency examination.

CHAPTER 2

Paragraph reference: 2.1

2.1.9.2
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011, Part 61 make no allowance for crediting of co-pilot flight time in aircraft certified for single pilot operation, and flown with a co-pilot, unless this is in a part 121, 135 or 127 operation, which requires 2 pilots (at night or IFR): therefore, in a corporate flight or private flight, even though operated as a multi-crew operation, no co-pilot time can be logged towards a higher licence.

2.1.9.4
South Africa’s regulation is more exacting in that the South African Civil Aviation Regulations, 2011, Part 61 makes provision for a student pilot attending an integrated course to be credited with PIC instrument time when flying under supervision provided all logbook entries are certified by the instructor supervising the flight.

Paragraph reference: 2.2

2.2.2.1
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011, 61.02.5(d) states that a student pilot is not allowed to fly solo on an international flight.

Paragraph reference: 2.3

2.3.2.2
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011, 61.10.1(2) state that a night rating is required to exercise the privileges of a licence by night. The night rating includes theoretical knowledge and instrument instruction as well as a skills test to be completed and that a night rating issued.

2.3.3.1.1
South Africa’s regulation is more exacting in that the South African Civil Aviation Regulations, 2011 indicates that an applicant for a Private Pilot License (A) must complete not less than 45 hours of total flight time in an aeroplane.

2.3.4.1.1
South Africa’s regulation is more exacting in that the South African Civil Aviation Regulations, 2011 indicates that an applicant for a Private Pilot License (H) must complete not less than 50 hours of total flight time in a helicopter.

2.3.4.1.2 South Africa’s regulation is more exacting in that the South African Civil Aviation Regulations, 2011 indicates that an applicant for a Private Pilot License (H) must complete not less than 15 hours of solo flight time in a helicopter.

Paragraph reference 2.4

2.4.6
South Africa has not implemented the licensing requirements for the issuance of an airship category rating.

Paragraph reference 2.6

2.6.4.1.1
South Africa’s regulation is more exacting in that the South African Civil Aviation Regulations, 2011 indicates an applicant for the issuance of an Airline Transport Pilot License (H) is required to have completed not less than 1500 hours flight time as a pilot of helicopters.

Paragraph reference 2.10

2.10.1.1
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011 indicates an applicant for the issuance of a free balloon pilot license shall not be less than 17 years old.

2.10.1.3.3 1
South Africa's regulation is more exacting in that the holder of a free balloon license cannot exercise their privileges of the license at night.

Paragraph reference: 2.9

2.9.1.5
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011, 61.00.7 requires that a commercial glider pilot have a Class 1 medical certificate and that a recreational glider pilot have a Class 2 medical certificate.

CHAPTER 4

Paragraph reference: 4.2

4.2.1.1
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011, 61.02.1 states that for aircraft maintenance engineers wishing to be issued with Category A, C, W & X rating age shall be 19 years or older and for Category B & D ratings shall be 21 years or older.

Paragraph reference: 4.3

4.3.1
Student air traffic controller not implemented. South African has implemented air traffic service assistant ratings. South African Civil Aviation Regulations, 2011, Part 65, subparts 3 -7.

Paragraph reference: 4.4

4.4.1.1
South Africa's regulation is different in character in that the South African Civil Aviation Regulations, 2011, 65.02.1(a) states that the applicant for an air traffic service licence

requires age to be not less than 18 years of age; a licence may be issued without ratings. Also see subparts 3 – 7.

Paragraph reference 4.6

4.6.1
South Africa does not license Flight Operations Officers or issue a flight dispatcher license.

2 Annex 2- Rules of the Air: Tenth Edition

CHAPTER 3

Paragraph reference: 3.6

3.6.2.2
South Africas Regulation is less protective in that the South African Civil Aviation Regulations, 2011. Regulation 91.03.4 9(12) (c) states that if the estimated time at the next applicable reporting point, flight information regional boundary, or aerodrome of intended landing, whichever comes first, is found to be in error in excess of three minutes from that notified to the responsible ATSU, a revised estimated time shall be notified to such ATSU as soon as possible; or... ICAO requires two minutes.

3 Annex 3 - Meteorological Services for International Air Navigation

CHAPTER 1. DEFINITIONS

Reference: 1.1

Aerodrome elevation. The elevation of the highest point of the landing area.

Remarks: Practiced but not yet transposed into national regulations.

ICAO meteorological information exchange model (IWXXM). A data model for representing aeronautical meteorological information.

Remarks: Practiced but not yet transposed into national regulations.

International airways volcano watch (IAVW). International arrangements for monitoring and providing warnings to aircraft of volcanic ash in the atmosphere.

Note. — The IAVW is based on the cooperation of aviation and non-aviation operational units using information derived from observing sources and networks that are provided by

States. The watch is coordinated by ICAO with the cooperation of other concerned international organizations.
Remarks: State is not a designated IAVW

CHAPTER 1. DEFINITIONS

Reference: 1.2

For the purpose of this Annex, the following terms are used with a limited meaning as indicated below:
a) to avoid confusion in respect of the term “service” between the meteorological service considered as an administrative entity and the service which is provided, “meteorological authority” is used for the former and “service” for the latter;
b) “provide” is used solely in connection with the provision of service;
c) “issue” is used solely in connection with cases where the obligation specifically extends to sending out the information to a user;
d) “make available” is used solely in connection with cases where the obligation ends with making the information accessible to a user; and
e) “supply” is used solely in connection with cases where either c) or d) applies.

Remarks: Practiced but not yet transposed into national regulations.

CHAPTER 2. GENERAL PROVISIONS

Paragraph reference: 2.2.7

Owing to the variability of meteorological elements in space and time, to limitations of observing techniques and to limitations caused by the definitions of some of the elements, the specific value of any of the elements given in a report shall be understood by the recipient to be the best approximation of the actual conditions at the time of observation.

Note. — Guidance on the operationally desirable accuracy of measurement or observation is given in Attachment A.

Remarks: Practiced but not yet transposed into national regulations.

Paragraph reference: 2.2.8

Owing to the variability of meteorological elements in space and time, to limitations of forecasting techniques and to limitations caused by the definitions of some of the elements, the specific value of any of the elements given in a forecast shall be understood by the recipient to be the most probable value which the element is likely to assume during the period of the forecast. Similarly, when the time of occurrence or change of an element is

given in a forecast, this time shall be understood to be the most probable time.

Note.— Guidance on the operationally desirable accuracy of forecasts is given in Attachment B.

Remarks: Practiced but not yet transposed into national regulations.

Paragraph reference: 2.2.9

The meteorological information supplied to the users listed in 2.1.2 shall be consistent with Human Factors principles and shall be in forms which require a minimum of interpretation by these users, as specified in the following chapters.

Note.— Guidance material on the application of Human Factors principles can be found in the Human Factors Training Manual (Doc 9683).

Remarks: Practiced but not yet transposed into national regulations.

CHAPTER 3. GLOBAL SYSTEMS, SUPPORTING CENTRES AND METEOROLOGICAL OFFICES

Paragraph reference: 3.3.3

The aerodromes for which landing forecasts are required shall be determined by regional air navigation agreement.

Remarks: Practiced but not yet transposed into national regulations.

CHAPTER 4. METEOROLOGICAL OBSERVATIONS AND REPORTS

Paragraph reference: 4.3.3

At aerodromes that are not operational throughout 24 hours in accordance with 4.3.1, METAR shall be issued prior to the aerodrome resuming operations in accordance with regional air navigation agreement.

Remarks: Practiced but not yet transposed into national regulations.

CHAPTER 5. AIRCRAFT OBSERVATIONS AND REPORTS

Paragraph reference: 5.3.4

In the case of the requirement to report during the climb-out phase, an aircraft shall be designated, at approximately hourly intervals, at each aerodrome to make routine observations in accordance with 5.3.1.

Remarks: Not yet transposed into national regulations.

Paragraph reference: 5.4

Aircraft not equipped with air-ground data link shall be exempted from making routine aircraft observations.

Remarks: Implemented in practice but not yet transposed into national regulations.

Paragraph reference: 5.6

When other meteorological conditions not listed under 5.5, e.g. wind shear, are encountered and which, in the opinion of the pilot-in-command, may affect the safety or markedly affect the efficiency of other aircraft operations, the pilot-in-command shall advise the appropriate air traffic services unit as soon as practicable.

Note. — Icing, turbulence and, to a large extent, wind shear are elements which, for the time being, cannot be satisfactorily observed from the ground and for which in most cases aircraft observations represent the only available evidence.

Remarks: Implemented in practice but not yet transposed into national regulations.

CHAPTER 6. FORECASTS

Paragraph reference: 6.5.2

When the density of traffic operating below flight level 100 warrants the issuance of AIRMET information in accordance with 7.2.1, area forecasts for such operations shall be prepared in a format as agreed between the meteorological authorities in the States concerned. When abbreviated plain language is used, the forecast shall be prepared as a GAMET area forecast, employing approved ICAO abbreviations and numerical values; when chart form is used, the forecast shall be prepared as a combination of forecasts of upper wind and upper-air temperature, and of SIGWX phenomena. The area forecasts shall be issued to cover the layer between the ground and flight level 100 (or up to flight level 150 in mountainous areas, or higher, where necessary) and shall contain information on en-route weather phenomena hazardous to low-level flights, in support of the issuance of AIRMET information, and additional information required by low-level flights.

Remarks: Practice but not yet transposed into national regulations.

CHAPTER 9. SERVICE FOR OPERATORS AND FLIGHT CREW MEMBERS

Paragraph reference: 9.1.2

Meteorological information supplied to operators and flight crew members shall cover the

flight in respect of time, altitude and geographical extent. Accordingly, the information shall relate to appropriate fixed times, or periods of time, and shall extend to the aerodrome of intended landing, also covering the meteorological conditions expected between the aerodrome of intended landing and alternate aerodromes designated by the operator.

Remarks: Practiced but not yet transposed into national regulations.

CHAPTER 9. SERVICE FOR OPERATORS AND FLIGHT CREW MEMBERS

Paragraph reference: 9.1.3

Meteorological information supplied to operators and flight crew members shall be up to date and include the following information, as agreed between the meteorological authority and the operators concerned:
a) forecasts of:
1) upper wind and upper-air temperature;
2) upper-air humidity;
3) geopotential altitude of flight levels;
4) flight level and temperature of tropopause;
5) direction, speed and flight level of maximum wind;
6) SIGWX phenomena; and
7) cumulonimbus clouds, icing and turbulence;

Note 1.— Forecasts of upper-air humidity and geopotential altitude of flight levels are used only in automatic flight planning and need not be displayed.

Note 2.— Forecasts of cumulonimbus clouds, icing and turbulence are intended to be processed and, if necessary, visualized according to the specific thresholds relevant to user operations.

b) METAR or SPECI (including trend forecasts as issued in accordance with regional air navigation agreement) for the aerodromes of departure and intended landing, and for take-off, en-route and destination alternate aerodromes;
c) TAF or amended TAF for the aerodromes of departure and intended landing, and for take-off, en-route and destination alternate aerodromes;
d) forecasts for take-off;
e) SIGMET information and appropriate special air-reports relevant to the whole route;

Note.— Appropriate special air-reports will be those not already used in the preparation of SIGMET.

f) volcanic ash and tropical cyclone advisory information relevant to the whole route;
g) as determined by regional air navigation agreement, GAMET area forecasts and/or area forecasts for low-level flights in chart form prepared in support of the issuance of AIRMET information, and AIRMET information for low-level flights relevant to the whole route;

h) aerodrome warnings for the local aerodrome;
i) meteorological satellite images;
j) ground-based weather radar information; and
k) space weather advisory information relevant to the whole route.

Remarks: Practiced but not yet transposed into national regulations.

Paragraph reference: 9.1.9

When necessary, the meteorological authority of the State providing service for operators and flight crew members shall initiate coordinating action with the meteorological authorities of other States with a view to obtaining from them the reports and/or forecasts required.

Remarks: Practiced but not yet transposed into national regulations.

Paragraph reference: 9.1.10

Meteorological information shall be supplied to operators and flight crew members at the location to be determined by the meteorological authority, after consultation with the operators concerned and at the time agreed between the aerodrome meteorological office and the operator concerned. The service for pre-flight planning shall be confined to flights originating within the territory of the State concerned. At an aerodrome without an aerodrome meteorological office at the aerodrome, arrangements for the supply of meteorological information shall be as agreed between the meteorological authority and the operator concerned.

Remarks: Practiced but not yet transposed into national regulations.

Paragraph reference: 9.2.1

Briefing and/or consultation shall be provided, on request, to flight crew members and/or other flight operations personnel. Its purpose shall be to supply the latest available information on existing and expected meteorological conditions along the route to be flown at the aerodrome of intended landing, alternate aerodromes and other aerodromes as relevant, either to explain and amplify the information contained in the flight documentation, or as agreed between the meteorological authority and the operator concerned, in lieu of flight documentation.

Remarks: Practiced but not yet transposed into national regulations.

CHAPTER 11. REQUIREMENTS FOR AND USE OF COMMUNICATIONS

Paragraph reference: 11.1.1

Suitable telecommunications facilities shall be made available to permit aerodrome meteorological offices and, as necessary, aeronautical meteorological stations to supply the required meteorological information to air traffic services units on the aerodromes for which those offices and stations are responsible, and in particular to aerodrome control towers, approach control units and the aeronautical telecommunications stations serving these aerodromes.

Remarks: Practiced but not yet transposed into national regulations.

Paragraph reference: 11.1.2

Suitable telecommunications facilities shall be made available to permit meteorological watch offices to supply the required meteorological information to air traffic services and search and rescue services units in respect of the flight information regions, control areas and search and rescue regions for which those offices are responsible, and in particular to flight information centres, area control centres and rescue coordination centres and the associated aeronautical telecommunications stations.

Remarks: Practiced but not yet transposed into national regulations.

Paragraph reference: 11.1.3

Suitable telecommunications facilities shall be made available to permit world area forecast centres to supply the required world area forecast system products to aerodrome meteorological offices, meteorological authorities and other users.

Remarks: Practiced but not yet transposed into national regulations.

Paragraph reference: 11.1.4

Telecommunications facilities between aerodrome meteorological offices and, as necessary, aeronautical meteorological stations and aerodrome control towers or approach control units shall permit communications by direct speech, the speed with which the communications can be established being such that the required points may normally be contacted within approximately 15 seconds.

Remarks: Practiced but not yet transposed into national regulations.

Paragraph reference: 11.1.8

Suitable telecommunications facilities shall be made available to permit meteorological

offices to exchange operational meteorological information with other meteorological offices.

Remarks: Practiced but not yet transposed into national regulations.

Paragraph reference: 11.2
Meteorological bulletins containing operational meteorological information to be transmitted via the aeronautical fixed service or the public Internet shall be originated by the appropriate meteorological office or aeronautical meteorological station.

Note. — Meteorological bulletins containing operational meteorological information authorized for transmission via the aeronautical fixed service are listed in Annex 10, Volume II, Chapter 4, together with the relevant priorities and priority indicators.

Remarks: Practiced but not yet transposed into national regulations.

Paragraph reference: 11.4

The content and format of meteorological information transmitted to aircraft and by aircraft shall be consistent with the provisions of this Annex.

Remarks: Practiced but not yet transposed into national regulations.

Paragraph reference: 11.5

D-VOLMET shall contain current METAR and SPECI, together with trend forecasts where available, TAF and SIGMET, special air-reports not covered by a SIGMET and, where available, AIRMET.

Note.— The requirement to provide METAR and SPECI may be met by the data link-flight information service (D-FIS) application entitled “Data link-aerodrome routine meteorological report (D-METAR) service”; the requirement to provide TAF may be met by the D-FIS application entitled “Data link-aerodrome forecast (D-TAF) service”; and the requirement to provide SIGMET and AIRMET messages may be met by the D-FIS application entitled “Data link-SIGMET (D-SIGMET) service”. The details of these data link services are specified in the Manual of Air Traffic Services Data Link Applications (Doc 9694).

Remarks: Practiced but not yet transposed into national regulations.

4 Annex 4 - Aeronautical Charts

CHAPTER 1. Definitions, Applicability and Availability

1.1 The following definitions are different from those published in the Annex:
Minimum Obstacle Clearance Altitude. The lowest altitude which may be used which will provide a minimum clearance of 1500ft (450m) above all objects located in an area contained within a sector of a circle 25NM (46km) radius centered on a radio aid to navigation, significant point, the Aerodrome Reference Point (ARP) or the Heliport Reference Point (HRP).

Prohibited Area. An airspace of defined dimensions, above the land areas or territorial waters of a State, within which the flight of civilian aircraft within the designated airspace is not permitted at any time under any circumstances.

CHAPTER 13. AERODROME/HELIPORT CHART —

13.5True and Magnetic North arrows and magnetic variation to the nearest degree and annual change of the magnetic variation shall be shown.

Partially implemented since November 2019.

13.6.1 This chart shall show: c) elevations and geoid undulations, to the nearest half-metre or foot, of the precision approach runway threshold, the geometric centre of the touchdown and lift-off area, and at the highest elevation of the touchdown zone of a precision approach runway;

Partially implemented since November 2019.

5 Annex 5 - Units of Measurement to be Used in Air and Ground Operations

NIL

6 Annex 6 - Operation of Aircraft

CHAPTER 6

PART 1

Paragraph reference: 6.2

6.2.2
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011 Part 91.04.13 (5) states that "no owner or operator of an aircraft used in general aviation operations for which the maximum certificated passenger seating is 20 or more on which is carried a cabin attendant shall operate the aircraft unless such aircraft is equipped with universal precaution kits." Technical standard

91.04.13 stipulates that such owner or operator shall ensure each aircraft carries on board at least two universal precaution kits. ICAO requires two only for aeroplanes authorized to carry more than 250 passengers.

Paragraph reference: 6.7

6.7.3

South Africa's regulation is less protective in that the South African Civil Aviation Regulations, 2011. Regulation 91.04.29 states "No person shall operate a pressurized aeroplane, for which the individual certificate of airworthiness was first issued on or after 1 January 1990, above 25 000 ft unless such aeroplane is equipped with a device to provide positive warning to the flight crew of any dangerous loss of pressurization." ICAO requires aeroplanes newly introduced into service on or after 1 July 1962.

6.7.4

South Africa's regulation is less protective in that the South African Civil Aviation Regulations, 2011. Regulation 91.04.29 states "No person shall operate a pressurized aeroplane, for which the individual certificate of airworthiness was first issued on or after 1 January 1990, above 25 000 ft unless such aeroplane is equipped with a device to provide positive warning to the flight crew of any dangerous loss of pressurization." ICAO requires the date to be after 1 July 1962.

Paragraph reference: 6.15

6.15.4

South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011. Regulation 121.05.5(3) All turbine-engined aeroplanes authorised under this Part (Part 121) to carry passengers shall be equipped with a TAWS which has a predictive terrain avoidance function.

ICAO only requires turbine powered aeroplanes above 5 700 kg to have forward looking (predictive) TAWS

South Africa's regulation is less protective in that the South African Civil Aviation Regulations, 2011. Regulation 135.05.5(1)(2)(3) All turbine-engine aeroplanes of a maximum certificated take-off mass in excess of 15 000 kg, for which the individual certificate of airworthiness is first issued on or after 1 July 1979, shall be equipped with TAWS.
(2) All turbine-engine aeroplanes of a maximum certificated take-off mass in excess of 5 700 kg or authorised to carry more than nine passengers for which the individual certificate of airworthiness is first issued on or after 1 January 2010, shall be equipped with a TAWS which has a predictive terrain avoidance functions.
(3) All turbine-engine aeroplanes of a maximum certificated take-off mass in excess of 5 700 kg or authorised to carry more than nine passengers, shall be equipped with a TAWS which has a predictive terrain avoidance function.

ICAO 6.15.5 Recommendation. - All turbine-engine aeroplanes of a maximum certificated take-off mass of 5 700 kg or less and authorised to carry more that five but not more than nine passengers should be equipped with a ground proximity warning system which provides the warnings of 6.15.8 a) and c), warning of unsafe terrain clearance and a forward looking terrain avoidance function.

Paragraph reference: 6.18

6.18.2
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011. Regulation 135.05.8(1). The operator of a large turbine-engine aeroplane shall not operate the aeroplane unless - (a) such aeroplane is equipped with a serviceable ACAS meeting ACAS II specifications, as prescribed in technical standard 91.04.31 of Document SA-CATS 91; and ICAO Annex 6 Part 1, 6.18.2 From 1 January 2005, all turbine-engined aeroplanes of a maximum certificated take-off mass in excess of 5 700 kg or authorised to carry more than 19 passengers shall be equipped with an airborne collision avoidance system (ACAS II)

SACAR 135.05.8 Says all aeroplanes above 5 700 kg regardless of passenger seats shall have ACAS II.

Paragraph reference: 6.18

6.18.3
South Africa's regulation is less protective in that the South African Civil Aviation Regulations, 2011. Regulation 91.04.28 Except as otherwise provided for in Part 121 and Part 135, no person may operate a turbine-engine aeroplane of a maximum certificated take-off mass in excess of 15 000 kg or authorised to carry more than 30 passengers, for which the individual certificate of airworthiness was first issued after 1 January 2007, unless such aeroplane is equipped with an ACAS that meets the specifications prescribed in Document SA-CATS 91,

ICAO recommends all aeroplanes be fitted with ACAS II.

CHAPTER 2

PART 2

Paragraph reference: 2.4.12

2.4.12.1
South Africa's regulation is less protective in that the South African Civil Aviation Regulations, 2011. Regulation 91.04.23 (3) (listed a-j) conditions exempt aeroplanes from having to carry a ELT, and 406 MHz ELT's are not mandatory. Whereas ICAO,

states in Annex 6 Part 2 2.4.12.1
Recommendation.— All aeroplanes should carry an automatic ELT.

CHAPTER 3

PART 2

Paragraph reference: 3.6.3.1.1.3
South Africa's regulation is less protective in that the South African Civil Aviation Regulations, 2011. Regulation 91.04.10(2)(b)states "No owner or operator shall operate an aircraft engaged in international general aviation operations which is an aeroplane with an MCM exceeding 5 700 kg for which the individual certificate of airworthiness was first issued on or after 1 January 2005 unless such aeroplane is equipped with a Type IA FDR that complies with the requirements prescribed in Document SA-CATS 91 whereas ICAO CoA first issued on or after 1 January 1989 should be equipped with a Type II FDR.

Paragraph reference: 3.6.10.3
South Africa's regulation is less protective in that the South African Civil Aviation Regulations, 2011.

Regulation 91.04.28 "Except as otherwise provided for in Part 121 and Part 135, no person may operate a turbine-engine aeroplane of a maximum certificated take-off mass in excess of 15 000 kg or authorised to carry more than 30 passengers, for which the individual certificate of airworthiness was first issued after 1 January 2007, unless such aeroplane is equipped with an ACAS that meets the specifications prescribed in Document SA-CATS 91, ICAO 3.6.10.3 Recommendation - All turbine-engine aeroplanes of a maximum certificated take-off mass in excess of 5 700 kg but not exceeding 1 500 kg, or authorised to carry more than 19 passengers, for which the individual airworthiness certificate is first issued after 1 January 2008, should be equipped with an airborne collision avoidance system (ACAS II).

CHAPTER 3

PART 3

Paragraph reference: 3.4

3.4.1
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011. Regulation 127.08.10 states "The operator of a Class 3 helicopter carrying passengers shall not operate such helicopter under IMC or above more than three eights of clouds within a radius of five nautical miles of the helicopter unless the latest weather reports or forecasts and any combination of them indicate that the weather along the planned route (including take-off and landing) with due regard for the provision of Regulation 127.08.12 allows flight under VFR an existing ceiling prescribed minimum heights established in terms of Regulation 127.07.6, and that the weather is focused to remain so until at least one hour after the estimated time of

arrival at the destination aerodrome or alternate aerodrome". In other words, South Africa does not allow Class 3 helicopter operations under IMC. Whereas in 3.4.1 ICAO allows for Class 3 helicopters operations under IMC to be conducted only over a surface environment.

CHAPTER 2

PART 2

Paragraph reference: 2.2.5

2.2.5.3
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations,2011. Regulation 12.02.1(1) states, "The PIC of an aircraft involved in an accident within the Republic, or if he or she is killed or incapacitated, a flight crew member, or if there are no surviving flight crew members or if they are incapacitated, the operator or owner, as the case may be, shall, as soon as possible but at least within 24 hours since the time of the accident, notify -
(a) the Director;
(b)An ATSU; or
(c) The nearest police station, of such accident.
ICAO standard 2.2.5.3 requires "the nearest appropriate authority" to be notified.

CHAPTER 2

PART 3

Paragraph reference: 2.6

2.6.2.1
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011. Regulation 91.7.7(8) requires the cloud base to be 400ft above the operating minima associated with instrument approach procedure. Meteorological conditions at the alternate will exist from two hours before to two hours after the estimated time of arrival. ICAO standard 2.6.2.1 requires the alternate heliport to be at or above the heliport operating minima.

CHAPTER 2

PART 3

Paragraph reference: 2.7

2.7.3
South Africa's regulation is more exacting in that the South African Civil Aviation Regulations, 2011. Regulation 91.7.7(9) requires "Suitable offshore alternates for helicopters may be subject to the following -
(a) The offshore alternates shall be used only after passing a PNR. Prior to a PNR, onshore alternates shall be used;

(b) Mechanical reliability of critical control systems and critical components shall be considered and taken into account when determining the suitability of the alternate;
(c)One-engine inoperative performance capability shall be attainable prior to arrival at the alternate;
(d) To the extent possible, deck availability shall be guaranteed; and
(e) Weather information must be reliable and accurate ICAO recommends offshore alternates not to be used when it is possible to carry enough fuel to have an onshore alternate. Offshore alternates should not be used in a hostile environment.

PANS - Training Doc 9868

NIL

7 Annex 7- Aircraft Nationality and Registration Marks

NIL

8 Annex 8- Airworthiness of Aircraft

CHAPTER 3
Paragraph 3.1

The Standards of this chapter are applicable in respect of all aircraft, except 3.3 and 3.4 which are not applicable in respect of all aircraft that are of a type of which the prototype was submitted to appropriate national authorities for certification before 13 June 1960.

RSA still issues Certificates of Airworthiness for aircraft that were type certificated prior to 13 June 1960, as long as it is still supported by the State of Design.

CHAPTER 4
Paragraph 4.2.1.5

The State of Design shall ensure that sensitive aviation security information is not transmitted when distributing mandatory continuing airworthiness information.

South Africa is yet to incorporate this annex provision into national regulations.

Paragraph 4.2.1.6

The State of Design shall ensure that sensitive aviation security information is securely transmitted to the appropriate authority in the States of Registry in accordance with Annex 17

South Africa is yet to incorporate this annex provision into national regulations.

Paragraph 4.2.4.3

The State of Registry shall ensure that sensitive aviation security information is not transmitted when distributing mandatory continuing airworthiness information.

South Africa is yet to incorporate this annex provision into national regulations.

Paragraph 4.2.4.4

The State of Registry shall ensure that sensitive aviation security information is securely transmitted to the appropriate authority in the State of Design in accordance with Annex 17

South Africa is yet to incorporate this annex provision into national regulations.

CHAPTER 2
Paragraph 2.2.3

Performance data shall be determined and scheduled in the flight manual so that its application by means of the operating rules to which the aeroplane is to be operated in accordance with 5.2 of Annex 6, Part I, will provide a safe relationship between the performance of the aeroplane and the aerodromes and routes on which it is capable of being operated. Performance data shall be determined and scheduled for the following stages for the ranges of mass, altitude or pressure-altitude, wind velocity, gradient of the take-off and landing surface for landplanes; water surface conditions, density of water and strength of current for seaplanes; and for any other operational variables for which the aeroplane is to be certificated

South Africa has adopted USA FAR Airworthiness design standards, which does not fully comply with ICAO standards.

CHAPTER 4
Paragraph 4.2

Special consideration shall be given to design features that affect the ability of the flight crew to maintain controlled flight.

South Africa has adopted USA FAR Airworthiness design standards, which do not fully comply with ICAO standards.

Paragraph 4.5

Electrical bonding and protection against lightning and static electricity shall be such as to:
a) protect the aeroplane, its systems, its occupants and those who come in contact with the aeroplane on the ground or water from the dangerous effects of lightning discharge and electrical shock; and
b) prevent dangerous accumulation of electrostatic charge.

South Africa has adopted USA FAR Airworthiness design standards, which do not fully comply with ICAO standards.

CHAPTER 6
Paragraph 6.4

The lights required by Annex 2 — Rules of the Air to be displayed by aeroplanes in flight or operating on the movement area of an aerodrome shall have intensities, colours, fields of coverage and other characteristics such that they furnish the pilot of another aircraft or personnel on the ground with as much time as possible for interpretation and for subsequent manoeuvre necessary to avoid a collision. In the design of such lights, due account shall be taken of the conditions under which they may reasonably be expected to perform these functions.

South Africa has adopted USA FAR Airworthiness design standards, which does not fully comply with ICAO standards.

9 Annex 9- Facilitation

CHAPTER 2

Paragraph 2.4

Contracting States shall not prevent an aircraft from calling at any international airport for public health reasons unless such action is taken in accordance with the International Health Regulations (2005) of the World Health Organization (WHO).

Partially Implemented - South Africa applies the World Health Regulations although not yet incorporated in Domestic Law.

Paragraph 2.43

Contracting States that require aircraft operators to apply for prior authorization shall:
a) establish procedures whereby such application will be dealt with promptly;
b) make such permission effective for a specific length of time or number of flights wherever possible; and
c) impose no fees, dues or charges for the issue of such permission.

More Exacting - There is prescribed application fee for commercial operations

CHAPTER 3

Paragraph 3.5

Contracting States shall ensure that no documents other than those provided for in this chapter shall be required by visitors for the entry into and departure from their territories.

More Exacting - South Africa requires a traveler who is a minor to produce a document showing the details of parents as well as a consent form from any of the parents not accompanying the minor.

CHAPTER 4

Paragraph 4.22

Contracting States that require documents for export clearance shall normally limit their requirement to a simplified export declaration.

Less Protective - Provision is made for simplified clearance in section 39D of the Customs and Excise Act, No.91 of 1964; however this provision is not yet effective

CHAPTER 8

Paragraph 8.47

Contracting States shall establish legislation, regulations and/or policies in support of assistance to aircraft accident victims and their families.

Less Protective – No Provision in place, under development

CHAPTER 9

Paragraph 9.1

Contracting States requiring the exchange of Advance Passenger Information (API)/ interactive API (iAPI) and/or Passenger Name Record (PNR) data from aircraft operators shall create a Passenger Data Single Window facility for each data category, or both data categories combined, that allows parties involved to lodge standardized information with a common data transmission entry point to fulfil all related passenger and crew data requirements for that jurisdiction.

Less Protective – No mechanism implemented yet

CHAPTER 10

Paragraph 10.1

In cases where proof of vaccination or prophylaxis is required by national authorities under the International Health Regulations (2005), Contracting States shall accept the International Certificate of Vaccination or Prophylaxis prescribed by the World Health Organization (WHO) in the International Health Regulations (2005).

Partially Implemented - South Africa fully implements the International Health Regulations although not yet incorporated in domestic law.

10 Annex 10- Aeronautical Telecommunications

Volume 1 - 3.1.3.9.3

South Africa uses a 2 letter International Morse Code and places the letter "I" as the third letter of the identification signal of ILS installation at various aerodromes.

Less Protective – Regulation amendment expected November 2023 which will result in a Different Means of Compliance filed.

11 Annex 11- Air Traffic Services

Chapter 2
Paragraph 2.11.3.3

An upper limit of a control area shall be established when either:
a) air traffic control service will not be provided above such upper limit; or
b) the control area is situated below an upper control area, in which case the upper limit shall coincide with the lower limit of the upper control area.

When established, such upper limit shall coincide with a VFR cruising level of the tables in Appendix 3 to Annex 2.

Done in practice but not included in regulations.

Chapter 3
paragraph 3.3.5.2

Where RCP/RSP specifications are applied, programmes shall be instituted for monitoring the performance of the infrastructure and the participating aircraft against the appropriate RCP and/or RSP specifications, to ensure that operations in the applicable airspace continue to meet safety objectives. The scope of monitoring programmes shall be adequate to evaluate communication and/or surveillance performance, as applicable.

Not implemented

Regional Supplementary Procedures - Doc 7030 PART 2 – COMMUNICATIONS

Paragraph 2.1 Under selected TMA’s in RSA (Special Rules Areas), the frequency 123.450 MHz is not applied. Paragraph reference: 3.1 PART 1 - RULES OF THE AIR, ATS

AND SAR Aircraft on VFR flights, and aircraft on IFR flights outside controlled airspace, are not required to maintain a watch on a radio station furnishing communications for the unit providing FIS in the FIR and are not required to file with that station information as to their position.

12 Annex 12- Search and Rescue

NIL

13 Annex 13- Aircraft Accident and Incident Investigation

NIL

14 Annex 14- Aerodromes

Volume 1

Chapter 1

Reference: Definition Arresting system. A system designed to decelerate an aeroplane overrunning the runway.

Mitigation: The State does not utilise arresting system. The State is implementing RESA requirements in the regulations.

Chapter 5

Paragraph 5.2.4.2

Recommendation. — A threshold marking should be provided at the threshold of a paved non-instrument runway where the code number is 3 or 4 and the runway is intended for use by other than international commercial air transport.

Mitigation: The State has adopted the standard to provide threshold marking on all paved runways of all codes.

Paragraph 5.2.4.3

A threshold marking should be provided, so far as practicable, at the thresholds of an unpaved runway. Note: The Aerodrome Design Manual (Doc 9157), Part 4, shows a form of marking which has been found satisfactory for the marking of downward slopes immediately before the threshold.

Mitigation: The State has adopted the standard on the use of edge markers to depict the start and end of unpaved runway.

Paragraph 5.2.6.5

On a non-precision approach runway where the code number is 2, an additional pair of touchdown zone marking stripes should be provided 150 m beyond the beginning of the aiming point marking.

Mitigation: The State does not have International Airports with Code 2 runways.

15 Annex 15- Aeronautical Information Services

NIL

16 Annex 16- Environmental Protection

NIL

17 Annex 17- Security

N/A

18 Annex 18- The Safe Transport of Dangerous Goods by Air

NIL

19 Annex 19- Safety Management

Chapter 3

Paragraph reference 3.4.2.

South Africa's State Safety Programme has not established the acceptable level of safety performance to be achieved.

Chapter 5

Paragraph reference: 5.3.5

Promotion of a positive safety culture has not yet formally started.

20 ICAO DOC 8168 - Aircraft Operations

Volume I- Flight Procedures

Part I Section 4 Chapter 2

Paragraph Reference: 2.3

2.3 MINIMUM SECTOR ALTITUDES (MSA)/TERMINAL ARRIVAL ALTITUDES (TAA)

Minimum sector altitudes are established for each aerodrome and provide at least 1500ft (450m) obstacle clearance within 46km (25NM) of the radio navigation aid associated with the approach procedure for that aerodrome.
RNAV based Minimum sector altitudes or Terminal arrival altitudes are established for each aerodrome and provide at least 1000 ft (300m) obstacle clearance within 46km (25 NM) of the significant point, the aerodrome reference point (ARP) or the heliport reference point (HRP) associated with the approach procedure for that aerodrome.

Volume II - Construction of visual and instrument flight procedures

Part I Section 4 Chapter 8

Paragraph Reference: 8.1.1

8.1.1 Minimum sector altitudes shall be established for each aerodrome where instrument approach procedures have been established. Each minimum sector altitude shall be calculated by:
a) taking the highest elevation in the sector concerned;
b) adding a clearance of at least:
i) 1500 ft (450m) for a Minimum sector altitude based on a radio navigation aid, or
ii) 1000 ft (300m) for a Minimum sector altitude based on a significant point, the aerodrome reference point (ARP) or the heliport reference point (HRP) (commonly referred to as an "RNAV MSA").
c) rounding the resulting value up to the next higher 100-ft increment, as appropriate.